



Module 1 challenge

1. Fill in the blank: If a test is statistically significant, the results are less likely to be due to _____ and more likely to be due to a real difference between the groups being compared.

1 / 1 point

- ☐ bias
- ☐ insufficient data
- ☐ causation
- ☒ random chance

✓ Correct

2. A company has 1,000 stores around the country. Each store tracks its inventory in live time using its point-of-sale system. Every night, each store's inventory data is sent to a central database. When customers go to the company's website, they can check if their local store has a particular item in stock, which the website does by querying the central database. Unfortunately, customers are coming to stores and finding that the item they want is sold out, even though the website said it was available. What data integrity problem does this scenario describe?

1 / 1 point

- ☐ Gathering
- ☒ Replication
- ☐ Transfer
- ☐ Manipulation

✓ Correct

3. In a survey about a new type of protein bar, 70% of respondents report they would buy the bar again. The margin of error for the survey is 4%. Based on that margin of error, what range reflects the population's true response?

1 / 1 point

- ☐ 74-78%
- ☒ 66-74%
- ☐ 66-70%
- ☐ 70-74%

✓ Correct

4. A car dealer conducts a survey to understand why customers choose their dealership. They are eager for positive feedback, so they email the survey to only those customers who purchased two or more vehicles from the dealership in the past five years. What is likely to result?

1 / 1 point

- ☒ Sampling bias
- ☐ Unbiased sampling
- ☐ Geographically limited sampling
- ☐ Random sampling

✓ Correct

5. Fill in the blank: To determine whether a survey or experiment has meaningful _____, a data team uses hypothesis testing.

1 / 1 point

- ☐ significance
- ☐ action items
- ☐ process steps
- ☒ results

✓ Correct

6. A data professional in the logistics industry wants to calculate the margin of error for a study about transportation route efficiency. They know the population size and sample size. What must they also know in order to accurately calculate margin of error?

1 / 1 point

- ☐ Testing methodology
- ☐ Distribution
- ☒ Confidence level
- ☐ Correlation

✓ Correct

7. A junior data analyst copies data from one computer to another over their company network. The network connection goes down during the process, which results in an incomplete copy of the data. What data integrity problem does this scenario describe?

1 / 1 point

- ☐ Manipulation
- ☒ Transfer
- ☐ Cleaning
- ☐ Replication

✓ Correct

8. Which of the following statements accurately describe sample size, population, and confidence level? Select all that apply.

0.8 / 1 point

- ☒ Confidence level deals with how strongly a data professional feels that they can rely on their sample reflecting the greater population.

✓ Correct

- ☐ Random sampling is a method that data professionals use to help address some of the issues with sampling bias.
- ☐ To be effective, most industries aim for a confidence level of at least 50-60%.
- ☒ Sample size is a part of a population that is representative of the population.

✓ Correct

✗ You didn't select all the correct answers